

Aaron Davis

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Charlotte, North Carolina

Education

North Carolina State University
Bachelor of Science in Computer Science

August 2023 – December 2025
Cumulative GPA: 3.798

Gaston College
Associate's of Science, Outstanding Graduate

May 2022 – May 2023
Cumulative GPA: 3.820

Experience

IT Developer Intern at Daimler Truck, Gastonia NC

May 2026 - Current

- Developing a full-stack Blazor/ASP.NET Core workforce management system to automate overtime allocation using business-rule logic based on employee skillset, seniority, and union compliance requirements.

Electrical Engineering Intern at Bosch Rexroth, Charlotte NC

3 Months

- Developed a fully automated testing procedure for a ball screw assembly, reducing manual testing time by 50%.
- Designed and implemented a GUI using Tkinter in Python to be used as a Human-Machine Interface for production testing.
- Performed routine testing and maintenance for the long term ball screw assembly test cabinet.

Calculus/Physics Tutor at Gaston College, Dallas NC

November 2022 – Current

- Responsible for facilitating learning, effective study habits, and the mastery of STEM concepts for Gaston College students.
- Consistently submitted tutoring session notes, timesheets, and other required records to supervisor in a timely manner.

Projects

IoT-Based Smart Blinds Automation System | *Python, MQTT, Raspberry Pi, Solidworks, NEMA 17 Stepper Motor*

- Designed and implemented a two-mode (automatic/manual) smart blinds solution leveraging Paho-MQTT library for real-time communication over AWS IoT Core, ensuring secure TLS/SSL communications.

Hitachi Energy Digital Twin Simulator | *C#, Git, Python, UML Design*

- Translating a thermo-chemical digital twin algorithm (Python) into C# and integrating it into the company codebase using Git/GitHub workflows.
- Reduced the runtime of a 60-year transformer simulation from 45 minutes to under 1 minute (97% improvement) through a new system design and numerical-solver optimization.
- Coordinated and facilitated sponsor meetings with Hitachi Energy, ensuring efficient use of time and clear next steps.

Technical Skills

Languages: Python, C#, Java, C, JavaScript, SQL

Frameworks: .Net Core, ASP.Net, Blazor, EF Core, Spring Boot, Node.js, React

Tools & DevOps: Git, GitHub Actions, Jenkins, JUnit, xUnit

Systems & IoT: MQTT, RabbitMQ, Raspberry Pi, Tkinter, AWS IoT Core